

WEEK 5

Aim:

To enhance the Week 4 webpage using JavaScript and to perform JavaScript form validation along with basic Node.js and NPM operations.

Description:

This experiment consists of two parts. In the first part, the Week 4 login dashboard is enhanced with a JavaScript `setTimeout()` callback function to redirect the user to the login page after 10 seconds of inactivity. In the second part, JavaScript-based HTML form validation is performed, followed by exploring Node.js installation, REPL, and NPM operations.

Part 1: Enhancement of Week 4 Experiment

Implement a 10-second inactivity timer using JavaScript.

Program Code:

1) Success page modified :

```
<html>
  <head>
    <title>Success-Page</title>
    <meta charset="UTF-8">
    <style>
      body{
        background-color: aliceblue;
        background-image: url("image.jpg");
      }

      .heading{
        height: 200px;
        align-items: center;
      }

      .heading img{
```

```
height: 100px;
width: 100px;
margin-left: 80px;
display: inline;
}
```

```
.text{
  align-items: center;
  display: flex;
  flex-direction: column;
  color: azure;
  position: absolute;
  top: 0px;
  left: 550px;
}
```

```
.navbar{
  height: 50px;
  display: flex;
  flex-direction: row;
  justify-content: space-around;
  background-color: purple;
}
```

```
a{
  color: white;
  text-decoration: none;
  margin-top: 10px;
}
```

```
.box{
  margin-top: 50px;
  height: 500px;
  width: 600px;
  margin-left: auto;
  margin-right: auto;
  background-color: lightblue;
}
```

```
    box-shadow: 10px 10px 10px grey;
    display: flex;
    flex-direction: column;
    justify-content: space-around;
}
```

```
.inner{
    background-color: aliceblue;
    border-left: 2px solid yellow;
    margin-left: auto;
    margin-right: auto;
}
```

```
li{
    margin-left: auto;
    margin-right: auto;
    color: blue;
    margin-left: 8px;
}
```

```
#greeting{
    color: blue;
}
```

```
#loginTime{
    color: blue;
}
```

```
</style>
```

```
</head>
```

```
<header>
```

```
<div class="heading">
```

```

```

```
<div class="text">
```

```
<h1>Anil Neerukonda Institute of Technology & Sciences</h1>
```

```
<h2>Campus Placement Portal</h2>
```

```
</div>
```

```

    </div>
</header>
    <div class="navbar">
        <a href="#">Home</a>
        <a href="#">About</a>
        <a href="#">Apply Job</a>
        <a href="#">UpComing Placements</a>
        <a href="#">Contact</a>
    </div>

<body>
    <div class="box">
        <h2 style="color: green; margin-left: 5px;">Login
Successful!</h2>
        <p style="color: darkblue;margin-left: 5px;">Welcome to ANITS
Central Placement Portal</p>
        <p id="loginTime"></p>
        <p id="greeting"></p>
    </div>
    <script>
        let now=new Date();
        let hours=now.getHours();
        let greet="";
        let time=setTimeout(timeUp,10000);

        function resetTimer(){
            clearTimeout(time);
            time=setTimeout(timeUp,10000);
        }

        window.addEventListener("click",resetTimer);
        window.addEventListener("mousedown",resetTimer);
        window.addEventListener("mouseup",resetTimer);
        window.addEventListener("mousedown",resetTimer);
        window.addEventListener("keydown",resetTimer);
        window.addEventListener("submit",resetTimer);
        window.addEventListener("keypress",resetTimer);
    </script>

```

```
function timeUp(){
    alert("Page redirected");
    window.location.href="index.html";
}

if(hours<12){
    greet="Good Morning!";
}

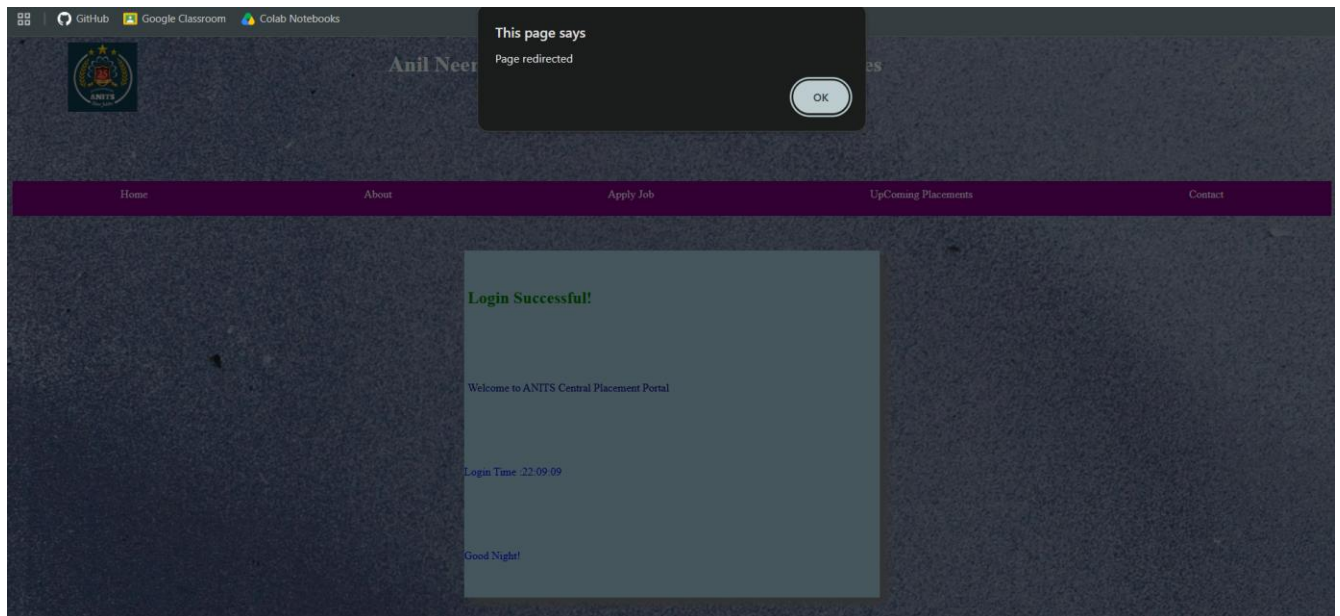
else if(hours<17){
    greet="Good Afternoon!";
}

else if(hours<21){
    greet="Good Evening!";
}

else{
    greet="Good Night!"
}

document.getElementById("greeting").innerHTML=greet;
document.getElementById("loginTime").innerHTML="Login
Time :"+now.toLocaleTimeString();
</script>
</body>
</html>
```

Output:



Part 2: JavaScript Form Validation and Node.js Basics

2.1 JavaScript-based HTML Form Validation.

Program Code:

```
<html>
<head><title> form validation</title>
</head>
<body>
<form onsubmit="return validateForm()">

  <label>Name:</label>
  <input type="text" id="name">
```

```
<br><br>
```

```
<label>Email:</label>
```

```
<input type="email" id="email">
```

```
<br><br>
```

```
<button type="submit">Submit</button>
```

```
</form>
```

```
<p id="error"></p>
```

```
<script> function
```

```
validateForm() {
```

```
    let name = document.getElementById("name").value.trim();
```

```
    let email = document.getElementById("email").value.trim();
```

```
    let error = document.getElementById("error");
```

```
    error.innerHTML = "";
```

```
    if (name === "") {        error.innerHTML
= "Name is required";
        return false;
    }
```

```
    if (email === "") {      error.innerHTML
= "Email is required";
        return false;
    }
```

```
    alert("Form submitted
successfully");  return true; }
</script>
</body>
</html>
```

Output:

Name:

Email:

 Please include an '@' in the email address. 'mohan7' is missing an '@'.

2.2 Node.js installation and verification, Node.js REPL practice, NPM – creation of package.json and installation of Express.

```
C:\Windows\System32\cmd.e  X  +  v

Microsoft Windows [Version 10.0.26200.9278]
(c) Microsoft Corporation. All rights reserved.

C:\Users\laksh\OneDrive\Desktop\fullstack\week5>node -v
v24.18.0

C:\Users\laksh\OneDrive\Desktop\fullstack\week5>npm -v
11.16.0

C:\Users\laksh\OneDrive\Desktop\fullstack\week5>node
Welcome to Node.js v24.18.0.
Type ".help" for more information.
> let a=10;
undefined
> let b=20;
undefined
> console.log(a+b);
30
undefined
> console.log("Successfully practiced in REPL");
Successfully practiced in REPL
undefined
> .exit;
Invalid REPL keyword
> .exit

C:\Users\laksh\OneDrive\Desktop\fullstack\week5>npm init -y
Wrote to C:\Users\laksh\OneDrive\Desktop\fullstack\week5\package.json:

{

C:\Users\laksh\OneDrive\Desktop\fullstack\week5>npm init -y
Wrote to C:\Users\laksh\OneDrive\Desktop\fullstack\week5\package.json:

{
  "name": "week5",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}

C:\Users\laksh\OneDrive\Desktop\fullstack\week5>npm install express
added 68 packages, and audited 69 packages in 3s

27 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

C:\Users\laksh\OneDrive\Desktop\fullstack\week5>
```

Details About Program:

Saved in: Week 5 folder of the repository.

Technologies used: HTML, CSS, JavaScript and Node.js.

Part 1 Execution: The enhanced HTML page is opened in a web browser.

Part 2 Execution: Form validation is executed in a web browser, while Node.js and NPM programs/commands are executed using the terminal or Command Prompt.

Working: Part 1 implements automatic redirection after 10 seconds of inactivity. Part 2 validates HTML form inputs and explores Node.js, REPL and NPM operations.

Result:

The Week 4 webpage was successfully enhanced with a 10-second inactivity timer. JavaScript form validation was performed, and Node.js, REPL and NPM operations were successfully explored.